

Problem set 2: Necessary conditions for the functioning of markets

Indifference curves and the Edgeworth box

1. Explain what is an indifference curve and why we usually assume that it is convex with respect to the origin.
2. Explain what is the “marginal rate of substitution”
3. Draw an indifference curve and a budget line. In your drawing, mark the point in which the utility is maximal and explain why that point is the relevant one.
4. Suppose that Max owns 5 bitcoins and 2 cars, whereas Julia owns 3 bitcoins and 10 cars.
 - a. Draw an Edgeworth box and mark the point that reflects this allocation.
 - b. Assume that both Max and Julia have a standard indifference curve. Draw these curves and mark the area that reflects Pareto efficiency.
5. Explain what is the “contract curve” and why it reflects Pareto efficient allocations.

Necessary conditions for trade

6. What are the two important features of a competitive equilibrium?
7. What is the first theorem of welfare economics? What is the second theorem of welfare economics? Explain what the second theorem adds to our understanding of allocative efficiency.
8. The government is thinking about taking some fixed amount of money from person A and giving it to person B. Suppose that the market is fully competitive. Person A is suing the government, claiming that this is “very inefficient”. Explain what that claim may be incorrect based on the second theorem of welfare economics.